

Sandeep Sohal

sandeepsohal.k@gmail.com | [\(647\)667-4744](tel:(647)667-4744) | <https://sandeepsohal.me/>
<https://github.com/sohal-sandy> | <https://www.linkedin.com/in/sandeep-sohal-a58404137/>

Education

Toronto Metropolitan University (formerly Ryerson University)

2020 – 2023

M.Sc. in Computer Science | Specialization in AI and ML | [Link to all courses](#)

CGPA: 3.96/4.33

Trent University

2015 – 2017

BBA Honours Program | Specialization in Information Systems & E-Commerce | [Link to all courses](#)

Dean's Honour Roll

University of Toronto (School of Continuing Studies)

Jan. 2025 - April 2025

Big Data Management Systems and Tools | Data Science Certification Course | [Course Description](#)

Experience

Academic Tutor

Oct. 2025 - Present

Megamind Learning Centre | Mississauga, ON

Math, English, Science, Python

- Tutor students from Grade 5 to high school in Math, English, Science, and Coding, including hands-on Python programming lessons tailored to individual learning needs.
- Simplify complex concepts and incentivize students to learn by using interactive activities, personalized feedback, and goal-oriented learning strategies.
- Assess student progress, identify learning gaps, and provide targeted support to strengthen understanding and improve academic performance.

Volunteer

June. 2025 - Aug. 2025

Squirrel Signs ASL | Toronto, ON

Python, SQL, GitHub, Azure, MediaPipe Holistic, CV

- Led a cross-functional team to build an MVP for a real-time ASL-to-English translation application by architecting an end-to-end deep learning pipeline using MediaPipe Holistic, sequence models (BiLSTM/Transformer), and a fine-tuned transformer language model (T5/BART).
- Managed dataset integration, preprocessing, and scalable data access (Azure Blob Storage) to enable efficient model development using the 2M-FLORES-ASL dataset.

Research Scientist

July. 2024 - Nov. 2024

Konnifel: Indian Institute of Technology-Madras | Remote

Python, SQL, GitHub, RdKit, DeepChem

- Worked under Dr. Sanjan TP Gupta's AI Lab to study large chemical datasets and build a chemical toxicity prediction project with **GCNs (Graphical Convolutional Neural Networks) and DL models**.
- Preprocessed **500,000+** molecular structures using RdKit and DeepChem.
- Web-scraped **over a million** chemical records from PubChem and MoleculeNet, expanding dataset coverage.

Data Engineer

March. 2024 - Nov. 2024

Alenka Media Inc. | Vancouver, BC | Remote

Python, SQL, GitHub, Project Management, Agile, Audio Analysis

- Worked with a team of 5 to develop an audio classification model with **90%** accuracy, categorizing genres, tempo, and language.
- Extracted over 200,000 audio features using Librosa, PyTorch, and TensorFlow.
- Generated spectrograms, mel-frequency cepstral coefficients (MFCCs) from MP3 and M4A audio files to extract relevant features from the files and reduced data pre-processing time by **40%** using optimized pipeline architecture.

Marketing Data Analyst

March. 2020 - Nov. 2023

Health Medica | Kitchener, ON

Python, SQL, Weka Data Mining Tool, Hootsuite, GitHub

- Increased online engagement by **35%** through data-driven digital marketing strategies.
- Forecast sales trends with **85%** accuracy using Weka and SQL-based analysis.

Research and Personal Projects

Interactive Web Application for Detection of Skin Conditions

Jan. 2025 - March. 2025

Computer Vision Project

OpenCV, CNN, PyTorch, Tensorflow

- Built a Flask web app to classify skin conditions using a CNN with transfer learning. Containerized with Docker, deployed on AWS (ECR & EC2), and integrated CI/CD with GitHub Actions.

A Study of Machine Learning and Deep Learning Models for Fake News Classification

Dec. 2021 - Jan. 2022

NLP project to identify reliable and unreliable news articles.

Python, Tensorflow, Google Colab GPU, BERT

- Processed **1M+** news articles, optimizing dataset handling for NLP applications and achieved **94% accuracy** on test data using the BERT model.

Technologies

Languages: Python, R, SQL, HTML/CSS, Java, JavaScript

Other: Git, GitHub, MongoDB, Tableau, Power BI, SQLite, Google Cloud Platform, AWS, Azure